

Problem-Solution Fit

Date	03 July 2026
Team ID	XXXXXX
Project Name	OptiCrop:Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviours, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] • Small-scale farmers • Large-scale farmers • Agricultural researchers • Agricultural officers • Policymakers	6.CUSTOMER LIMITATIONS [CL] • Limited technical knowledge • Limited internet access in rural areas • Budget constraints • Lack of regular soil testing • Dependence on local advice	5. AVAILABLE SOLUTIONS (PROS & CONS) [AS] Existing Solutions • Advice from experienced farmers • Agriculture officers • Internet articles and videos Pros • Easy to access • Low cost • Based on practical experience Cons • Advice may differ from person to person
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		• Not personalized for each farm • Less accurate under changing weather conditions
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] • Difficulty selecting the right crop every farming season • Uncertain weather affecting crop decisions • Lack of knowledge about soil nutrients • High farming costs due to wrong crop selection • Dependence on traditional farming methods	9. PROBLEM ROOT/ CAUSE [RC] • Lack of accurate crop selection guidance • Changing climate conditions • Limited awareness of soil health • Traditional decision-making practices • Insufficient access to agricultural information	7. BEHAVIOR + ITS INTENSITY [BE] • Frequently consults neighboring farmers • Checks weather before planting • Uses previous farming experience • Tries to reduce farming risks • Searches for better farming methods
3. TRIGGERS TO ACT [TR] • New farming season begins • Poor harvest in the previous season • Soil test results become available • Changes in weather conditions • Need to increase crop yield and income	10. YOUR SOLUTION [SL] • Recommends the best crop based on soil conditions. • Analyzes N, P, K, pH, and weather data. • Helps farmers choose suitable crops with confidence.	8. CHANNELS OF BEHAVIOR [CH] Online • Agricultural websites • YouTube farming channels • Weather applications • Government agriculture portals

	• Improves crop yield through better planning. • Reduces losses caused by wrong crop selection.	
4. EMOTIONS (BEFORE / AFTER) [EM] Before • Confused • Worried • Uncertain • Stressed After • Confident • Satisfied • Hopeful • Motivated	• Optimizes the use of water and fertilizers. • Provides quick and accurate crop recommendations. • Supports sustainable and efficient farming practices.	Offline • Agriculture officers • Soil testing centers • Local fertilizer shops • Farmer meetings • Neighboring farmers